Exercise PlantUML Code

（1）Employment by enterprises

```
@startuml


class Job {

}


class Enterprise {

    - Name: string

    - Address: string

    - Phone: string

    - Brief_introduction: string

}


class Personnel {

    - Name: string

    - Age: int

    - Gender: string
```

```
    - Address: string
    - Phone: string
    - Resume: string
}

class Employment {
    - StartDate: date
    - EndDate: date
    - Brief_description: string
- Position: string
}

class EmploymentContract {
}

Enterprise    "1" -- "*" Personnel
(Enterprise, Personnel) .. Employment
Employment "1" -- "1" EmploymentContract
```

```
Personnel "*" --> "1" Job

Enterprise o-- Job


@enduml
```

（2）Hotel room business

```
@startuml


class Staff {

    - Staffno: string

    - Name: string

    - Gender: string

    - PhoneNumber: string

    - Position: string

}


class Booker {

    - Name: string
```

```
    - Gender: string
    - ID_Card: string
    - PhoneNumber: string
}

class Reservation {
    - ReservationNo: string
    - BookingDate: date
    - BookingRoomNo: int
    - CheckinDate: date
    - BookingDays: int
    - Deposit: double
}

class CheckInRecord {
    - CheckInNo: string
    - CheckInDate: date
    - CheckInDays: int
```

```
    - Deposit: double
}

class Guest {
    - Name: string
    - Gender: string
    - Phone: string
    - ID_Card: string
}

class Room {
    - RoomNo: string
    - Floor: string
    - RoomType: string
    - Price: int
    - RoomStatus: string
}
```

```
Booker "1" -- "*" Reservation

Reservation "*" -- "1" Room

CheckInRecord "*" -- "1" Guest

Room "1" -- "*" CheckInRecord

Staff "1" -- "*" Reservation

Staff "1" -- "*" CheckInRecord


@enduml
```

（3）Project management system

```
@startuml


class "Work Product" {

        - Percent Complete: double

        - Description: string

        + Validate()

}
```

```
class Requirement {
    - Media: string
    + Validate()
    + Publish()
}

class System {
    - Platform
    + Validate()
    + Deploy()
}

class Project {
    - Name
    - StartDate
    - EndDate
}
```

```
class Manager {
    - Name
    - PhoneNumber
    + Initiate Project()
    + Terminate Project()
}

class Team {
    - Description
}

Requirement --|> "Work Product"
System --|> "Work Product"

Project "1" -- "*" Requirement

Project "1" -- "*" System
```

```
Manager "1" -- "*" Project

Manager    -- Team

Team "1" -- "*" Project

@enduml
```

（4）Film shooting software program

```
@startuml

class Scene {
        - Code: String
        - Description: Text
}

class Internal {
```

```
    - Theater: String
}

class External {
    - Night_scene: Boolean
}

class Setup {
    - Code: String
    - Photographic_parameters: Text
}

class Take {
    - Number: Integer
    - Filmed_meters: Real
    - Reel: String
}
```

```
class Location {
    - Name: String
    - Address: String
    - Description: Text
}

Internal --|> Scene : (complete, disjoint)
External --|> Scene : (complete, disjoint)

Scene "1" -- "1..*" Setup : stp_for_scn

Setup "1" -- "1..*" Take : tk_stp

External "0..*" -- "1" Location : located

@enduml
```

（5）Document system

```
@startuml

class Document {
    + PageNumber
    + Open()
    + Save()
    + Print()
    + New()
}

class Page {
    - PageNumber
    + NewPage()
    + HideHeader()
    + HideFooter()
    + InsertPicture()
```

```
        + InsertTable()
}

class Trimming {
        - Date
        - Time
        - PageNumber
        + Display()
        + Change()
        + Hide()
}

class Character {
        - ASCIIcode
        - Type
        + Normal()
        + Italic()
        + Bold()
```

```
        + Underline()
}

class Table {
        - Rows
        - Columns
        + InsertRow()
        + InsertColumn()
        + NewTable()
        + InsertPicture()
}

class Picture {
        - ImageType
        - ImageSize
        + Insert()
        + Delete()
}
```

```
class Cell {
        - CellFormat
        + Edit()
}

Document "1" *-- "1..*" Page

Page "1" o-- "0..1" Trimming

Page "1" -- "*" Character

Page "1" -- "0..*" Table

Page "1" -- "0..*" Picture

Table "1" *-- "1..*" Cell
```

```
Cell "1" o-- "0..*" Character

Cell "1" o-- "0..*" Picture

@enduml
```

（6）Hospital visit process system

```
@startuml

class FamilyInsured {
    - Name
    - Address
    - Telephone
    - InsuranceCarrier
    - PolicyNumber
}
```

```
class Patient {
    - Name
    - Gender
    - BirthDate
    - LastVisitDate
    - Allergies
    - Notes
}

class Appointment {
    - Date
    - Time
    - Reason
}

class Doctor {
    - Name
    - Specialty
```

- Address
- Telephone
- PagerNumber
- SSN

}

class Visit {

- Date
- Weight
- BloodPressure
- Temperature
- Reason
- DiagnosisNotes
- TreatmentNotes

}

class Prescription {

- Date

- Medication

- Dosage

- NumberRe

}

FamilyInsured "1" -- "1..*" Patient

FamilyInsured "1..*" -- "1" Doctor

Patient "1" -- "0..*" Appointment

Patient "1" -- "0..*" Visit

Visit "0.." -- "0..1" Appointment

Doctor "1" -- "1..*" Visit

Visit "1" -- "0..*" Prescription

@enduml

（7）Movie Store System

@startuml

class User {

- FirstName

- LastName

}

class Subscriber {

- PhoneNumber

- Address

}

```
class Card {
    - AccountBalance
    - Code
}

class "Movie-Shop" {
    + ListMovies()
    + Order()
    + Subscribe()
}

class Movie {
    - Name
    - Type
}

class "Movie-Rent" {
    - State: (in, out)
```

```
        - RentDate
        - Price
}

class "Movie-Buy" {
        - State
        - Price
}

class Order {
        - Date
        - User
}

Subscriber --|> User

User "*" -- "1" "Movie-Shop"
```

Subscriber "1" -- "1" Card

Subscriber "*" -- "1..*" "Movie-Rent"

"Movie-Shop" "1" -- "*" Card

"Movie-Shop" "1" -- "*" Order

Movie <|-- "Movie-Rent"

Movie <|-- "Movie-Buy"

Movie "*" --o "1" "Movie-Shop"

"Movie-Buy" "1..*" -- User

Order "*" -- "1..*" "Movie-Buy"

@enduml

（8）Aviation management system

@startuml

class Airline {

- ID

- Name

}

class Flight {

- ID

- ArrivalTime

- DepartureTime

}

```
class Airport {
    - ID
    - Name
}


class Aircraft {
    - Name
    - Repair
    - Landed
}


class "Aircraft Type" {
    - Name
}
```

```
class Pilot {
        - Name
}


class Pilot1
class Pilot2
class Pilot3

Airline "1" -- "*" Flight

Airline "*" -- "*" Aircraft

Flight "*" -- "1" Airport

Flight "*" -- "1" Aircraft
```

```
Flight    -- "2..n" Pilot

Aircraft "*" -- "1" "Aircraft Type"

"Aircraft Type" "*" -- "1..n" Pilot

Pilot3 "1" -- "*" "Aircraft Type"

Pilot <|-- Pilot1
Pilot <|-- Pilot2
Pilot <|-- Pilot3

@enduml
```

（9）Bank account system

```
@startuml
```

```
class Customer {
    - Name : String
    - Address : String
}

class Account {
    - CashBalance : Double
}

class InvestmentAccount

class SavingAccount {
    - InterestRate : Double
}

class Stock {
    - Ticker : String
```

```
        - Quantity : Integer
}

class StockOrder {
        - Commission : Double
}

Customer "1" -- "0..*" Account

Account <|-- InvestmentAccount
Account <|-- SavingAccount

InvestmentAccount    -- "0..*" Stock

Stock -- StockOrder

@enduml
```

（10）Restaurant catering system

```
@startuml

class Ingredient {
    - Name
    - Unit
    - Stock
}

class Dish {
    - ID
    - Name
}

class Quantity
```

```
class TimesServed


class Meal {
    - Date
    - StartTime
    - EndTime
}


class Person {
    - Name
    - Address
}


class Client {
    - Tax_ID
}


class Waiter {
```

```
        - ID
        - Phone
}

class Table {
        - ID
        - Capacity
}

Ingredient "1..*" -- "*" Dish

(Ingredient, Dish) .. Quantity

Dish "*" -- "*" Meal

(Dish, Meal) .. TimesServed

Meal "*" -- "1" Table
```

```
Meal "*" -- "0..1" Client

Meal "*" -- "1" Waiter

Person <|-- Client
Person <|-- Waiter

@enduml
```

（11）Supplier order system

```
@startuml

class Customer {
    - ID: String
    - Name: String
```

```
        - ContactInformation: String
        + getCustomerInfo(): String
        + placeOrder(order: Order, supplier: Supplier):
boolean
    }



    class Supplier {
        - ID: String
        - Name: String
        - contactInformation: String
        + getSupplierInfo(): String
        + receiveOrder(order: Order): boolean
        + createDelivery(order: Order): Delivery
    }



    class Order {
```

```
    - ID: String
    - orderDate: Date
    - orderStatus: String
    + getOrderInfo(): String
    + addOrderLine(orderLine: OrderLine): boolean
}


class OrderLine {
    - ID: String
    - orderQuantity: int
    + getOrderLineInfo(): String
}


class Part {
    - ID: String
    - Name: String
```

```
}



class Delivery {
    - ID: String
    - Date: Date
    - Status: String
    + getDeliveryInfo(): String
    + addPart(part: Part, quantity: int): boolean
    + bindOrder(order: Order): void
}

Customer "1" -- "*" Order

Supplier "1" -- "*" Order

Supplier "1" -- "*" Part
```

Order "1" *-- "*" OrderLine

OrderLine "1" -- "1" Part

Supplier "1" -- "*" Delivery

Delivery "1" -- "1" Order

Delivery "1" -- "*" Part

@enduml

（12）The banking system of the United Kingdom

class Bank {

- Name: String

+ OpenBranch(branchCode: String, address: String): Branch

```
        + GetBranch(branchCode: String): Branch
    }

    class Branch {
        - Code: String
        - Address: String
        + OpenAccount(accountNo: String, balance: double): Account
        + GetAccount(accountNo: String): Account
    }

    class Account {
        - AccountNumber: String
        - Balance: double
        + Deposit(amount: double): void
        + Withdraw(amount: double): boolean
        + GetBalance(): double
    }
```

```
class Cheque {
    - ChequeNumber: String
    - Amount: double
    - IssueDate: Date
    - IsCashed: boolean
    + cash(): boolean
}

Bank "1" -- "*" Branch

Branch "1" -- "*" Account

Account "0..1" -- "*" Cheque
```

（13）Programming language syntax system

```
@startuml

class Module {
    - Name: String
    + GetModuleInfo(): String
    + AddFeature(feature: Feature): boolean
}

class Feature {
    - Name: String
    + GetFeatureInfo(): String
}

class Variable {
    - Type: String
    - Value: Object
    + GetVariableInfo(): String
}
```

```
class Routine {
    - Name: String
    - Type: String
    + GetRoutineInfo(): String
}

class NestedModule {
    - Name: String
    + GetNestedModuleInfo(): String
}

class DeclarationPart {
    + AddLocalFeature(feature: Feature): boolean
    + GetLocalFeatures(): Feature[]
}

class StatementPart {
```

```
    + AddStatement(statement: Statement): boolean
    + GetStatements(): Statement[]
}

class Statement {
    + Execute(): void
}

class Loop {
    - LoopCondition: String
    - LoopBody: Statement[]
    + GetLoopInfo(): String
}

class Conditional {
    - Condition: String
    - IfBody: Statement[]
    - ElseBody: Statement[]
```

```
        + GetConditionalInfo(): String
}

class Assignment {
        + GetAssignmentInfo(): String
}



Module "1" *-- "*" Feature
Variable --|> Feature
Routine --|> Feature
NestedModule --|> Feature

Routine "1" *-- "1" DeclarationPart
Routine "1" *-- "1" StatementPart

StatementPart "1" *-- "1..*" Statement
```

```
Loop --|> Statement

Conditional --|> Statement

Assignment --|> Statement


DeclarationPart "1" *-- "*" Feature


Assignment "1" -- "1" Variable


@enduml
```

（14）Electronic Workbench System

```
@startuml


class Document {

        - ID: String

        - Name: String

        - Content: String
```

```
    - CreateTime: Date
    + GetDocumentInfo(): String
}

class ElectronicDesk {
    - ID: String
    - UserName: String
    + GetElectronicDeskInfo(): String
}

class BlottingPad {
    + ProvideWordProcessing(): void
    + PlaceDocument(doc: Document): boolean
    + RemoveDocument(): Document
}

class FilingCabinet {
    - ID: String
```

```
    + GetFilingCabinetInfo(): String
}

class Drawer {
    - ID: String
    - Name: String
    + GetDrawerInfo(): String
    + StoreDocumentDirectly(doc: Document): boolean
}

class Folder {
    - Name: String
    + GetFolderInfo(): String
    + StoreDocument(doc: Document): boolean
}

class MailService {
    - ID: String
```

```
        + AutoDeliverToInTray(doc: Document, targetDesk:
ElectronicDesk): boolean
        + AutoSendFromOutTray(): boolean
    }

    class MailTray {
        - Type: String
        - Status: String
        + GetMailTrayInfo(): String
        + AddDocument(doc: Document): boolean
        + RemoveDocument(doc: Document): boolean
    }

    ElectronicDesk "1" *-- "1" BlottingPad
    ElectronicDesk "1" *-- "1" FilingCabinet
    ElectronicDesk "1" *-- "1" MailService

    MailService "1" *-- "3" MailTray
```

FilingCabinet "1" *-- "*" Drawer

Drawer "1" *-- "*" Folder

BlottingPad "1" -- "0..1" Document

Drawer "1" *-- "*" Document

Folder "1" -- "*" Document

MailTray "1" -- "*" Document

@enduml

（15）The Emacs editor system

```
@startuml

class InfoSystem {
    - ID: String
    - StartTime: Date
    + GetSystemInfo(): String
    + GetDirectoryNode(): InfoNode
    + AddInfoFile(file: InfoFile): boolean
}

class File {
    - Path: String
    - Name: String
    - Description: String
    + GetFileInfo(): String
    + AddInfoNode(node: InfoNode): boolean
}
```

```
class Node {
    - ID: String
    - Title: String
    - Content: String
    - IsDirectoryNode: boolean
    + GetNodeInfo(): String
    + AddLink(link: Link): boolean
}

class Link {
    - ID: String
    + GetLinkInfo(): String
    + GetTargetNode(): InfoNode
}

class UpLink {
    + GetUpLinkInfo(): String
}
```

```
class NextLink {
    + GetNextLinkInfo(): String
}

class PreviousLink {
    + GetPreviousLinkInfo(): String
}

class Menu {
    - Name: String
    + GetMenuInfo(): String
    + AddMenuEntry(entry: MenuEntry): boolean
}

class MenuEntry {
    - Name: String
    - Description: String
```

```
    + GetMenuEntryInfo(): String
    + SetTargetNode(node: InfoNode): void
}

class HistoryList {
    - MaxCapacity: int
    - RecordList: List<HistoryRecord>
    + GetHistoryInfo(): String
    + AddHistoryRecord(record: HistoryRecord): boolean
    + RetracePath(): List<InfoNode>
}

class HistoryRecord {
    - Time: Date
    + GetHistoryRecordInfo(): String
}

InfoSystem "1" *-- "*" File
```

File "1" *-- "*" Node

InfoSystem "1" -- "1" Node

Node "1" *-- "*" Link

UpLink --|> Link

NextLink --|> Link

PreviousLink --|> Link

Node "1" *-- "0..1" Menu

Menu "1" *-- "*" MenuEntry

MenuEntry "1" -- "1" Node

InfoSystem "1" *-- "1" HistoryList

```
HistoryList "1" *-- "*" HistoryRecord

HistoryRecord "1" -- "1" Node

@enduml
```

（16）Window Manager System

```
@startuml

class WindowManager {
    - Desktop: Desktop
    + StartSession(): void
    + EndSession(): void
}

class Desktop {
```

```
    + GetApplications(): List<Application>
    + AddApplication(app: Application): void
    + GetWindows(): List<Window>
}

class Application {
    - Name: String
    - IsRunning: boolean
    - Icon: Icon
    + Start(): void
    + Terminate(): void
    + GetWindows(): List<Window>
}

class ControlApplication{
    + Terminate(): void
}
```

```
class Icon {
    - Image: String
    - AssociatedApp: Application
}

class Window {
    - Title: String
    - TitleBar: TitleBar
    - MaximizeBtn: Button
    - MinimizeBtn: Button
    - ControlMenuBox: ControlMenuBox
    - HorizontalScrollbar: Scrollbar
    - VerticalScrollbar: Scrollbar
    + Maximize(): void
    + Minimize(): void
}

class ApplicationWindow {
```

```
    - MenuBar: MenuBar
    + GetDocumentWindows(): List<DocumentWindow>
    + AddDocumentWindow(window: DocumentWindow): void
}

class DocumentWindow {
    - ParentApplicationWindow: ApplicationWindow
}

class TitleBar {
    - Title: String
}

class MenuBar {
    - Menus: List<Menu>
    + GetMenu(name: String): Menu
}
```

```
WindowManager "1" *-- "1" Desktop

Desktop "1" *-- "*" Application

Desktop "1" *-- "*" Window


Application "1" *-- "1" Icon

Application "1" *-- "*" Window

ControlApplication --|> Application


Window "1" *-- "1" TitleBar

Window "1" *-- "1" maximizeBtn: Button

Window "1" *-- "1" minimizeBtn: Button

Window "1" *-- "1" controlMenuBox: ControlMenuBox

Window "0..1" *-- "1" horizontalScrollbar: Scrollbar

Window "0..1" *-- "1" verticalScrollbar: Scrollbar


ApplicationWindow "1" *-- "1" MenuBar

ApplicationWindow "1" *-- "*" DocumentWindow
```

```
ApplicationWindow --|> Window

DocumentWindow --|> Window


@enduml
```

（17）Electronic Bulletin Board System

```
@startuml


class ElectronicNoticeboard {

    + Login(user: User): void

    + Logout(user: User): void

    + GetWorkspaces(): List<UserWorkspace>

}


class User {

    - Name: String
```

```
    - IsRegistered: boolean
    + GetPrivateNotices(): List<Notice>
    + RemoveFromPrivateView(notice: Notice): void
}

class UserWorkspace {
    - ActiveArea: String
    + WwitchArea(area: String): void
}

class NoticeBoard {
    + GetNotices(): List<Notice>
    + AddNotice(notice: Notice): void
}

class DiscussionGroupsArea {
    + GetDiscussionGroups(): List<DiscussionGroup>
    + AddDiscussionGroup(group: DiscussionGroup): void
```

```
}

class Notice {
    - Title: String
    - Content: String
    - Date: Date
    - CreationDate: Date
    - IsArchived: boolean
    + IsExpired(): boolean
}

class DiscussionGroup {
    - Topic: String
    - Creator: User
    - Discussions: List<Discussion>
    + GetAccessPermissions(user: User): String
    + SetPermission(user: User, permission: String): void
}
```

```
class Discussion {
    - InitialPost: Post
    - DerivedFrom: Discussion
    + GetPosts(): List<Post>
    + AddPost(post: Post): void
}

class Post {
    - Content: String
    - Author: User
    - CreationDate: Date
    - ParentPost: Post
    - IsRead: boolean
    + GetReplies(): List<Post>
}

class Archive {
```

```
    - Notice: List<Notice>
    - Post: List<Post>
    + AddNotice(notice: Notice): void
    + AddPost(post: Post): void
}

ElectronicNoticeboard "1" *-- "*" UserWorkspace
ElectronicNoticeboard "1" *-- "1" Archive
ElectronicNoticeboard "1" -- "*" User

UserWorkspace "1" *-- "1" NoticeBoard
UserWorkspace "1" *-- "1" DiscussionGroupsArea
UserWorkspace "1" -- "1" User

NoticeBoard "1" *-- "*" Notice
Notice "1" -- "0..*" User

DiscussionGroupsArea "1" *-- "*" DiscussionGroup
```

```
DiscussionGroup "1" *-- "*" Discussion

DiscussionGroup "1" -- "*" User


Discussion "1" *-- "*" Post


Post "1" -- "1" User


Archive "1" -- "*" Notice

Archive "1" -- "*" Post


@enduml
```

（18）Library system

```
@startuml


class Library {
```

```
    - Name: String
    - Address: String
    - OpeningHours: String
    + GetLibraryInfo(): String
    + AddReader(reader: Reader): boolean
    + UpdateCatalogue(book: Book): boolean
}


class BookCatalogue {
    - ID: String
    - UpdateTime: Date
    + GetCatalogueInfo(): String
    + AddBook(book: Book): boolean
    + SearchBookByISBN(isbn: String): Book
    + SearchBookByTitle(title: String): List<Book>
}
```

```
class Book {
    - Title: String
    - AuthorName: String
    - ISBN: String
    + GetBookInfo(): String
    + AddBookCopy(copy: BookCopy): boolean
}

class BookCopy {
    - AccessionNumber: String
    - BookStatus: String
    + GetBookCopyInfo(): String
    + IsAvailable(): boolean
}

class Reader {
    - ID: String
```

```
        - Name: String
        - Address: String
        - TicketCount: int
        + GetReaderInfo(): String
        + BorrowBook(copy: BookCopy, returnDate: Date): BorrowRecord
        + ReturnBook(copy: BookCopy): boolean
    }

    class BorrowRecord {
        - ID: String
        - BorrowDate: Date
        - DueReturnDate: Date
        + GetBorrowRecordInfo(): String
        + IsOverdue(): boolean
    }

    Library "1" *-- "1" BookCatalogue
```

BookCatalogue "1" *-- "*" Book

Book "1" *-- "*" BookCopy

Library "1" *-- "*" Reader

Reader "1" -- "*" BorrowRecord

BookCopy "1" -- "*" BorrowRecord

@enduml

（19）Veterinary Clinic System

@startuml

class Animal {

```
        - ID
        - Name
}

class Breed {
        - Name
}

class Condition {
        - Common_Name
        - SCI_Name
}

class Appointment {
        - Date
        - Time
}
```

```
class Person {
    - ID
    - Name
    - Address
    - Phone
}

class Owner

class Physician

Animal "*" -- "1" Breed

Breed "*" -- "*" Condition

Animal "1" -- "*" Appointment

Appointment "*" -- "*" Condition
```

```
Animal "1..*" -- "0..1" Owner


Person <|-- Owner

Person <|-- Physician



Appointment "*" -- "1" Physician


@enduml
```

（20）Prepaid mobile phone

```
@startuml


class Contract {

        + GetCharges(): double

        + GetBalance(): double
```

```
        + SetBalance(double)
}

class "Basic Contract" {
        - Balance: double
        - ContractNumber: int
}

class Option

class Client

class Data
class SMS
class Phone
class "Double Transfer"

"Basic Contract" ..|> Contract
```

```
Option ..|> Contract


Option <|-- Data

Option <|-- SMS

Option <|-- Phone

Option <|-- "Double Transfer"


Client -- Contract


@enduml
```

（21）Archives management system

```
@startuml


class ArchiveManagementSystem {

    + Login(user: User): boolean
```

```
    + BackupData(conditions: BackupConditions): BackupFile
    + RestoreData(backupFile: BackupFile): void
}

class User {
    - UserName: String
    - Name: String
    - Department: String
    - Address: String
    - Password: String
    - DisplayRowsPerPage: int
    - IsDeleted: boolean
    + ModifyPassword(newPassword: String): void
    + ModifyDisplayRows(rows: int): void
}

class AdminUser {
```

```
    + AddUser(user: User): void
    + DeleteUser(username: String): void
    + ModifyUserAttributes(user: User): void
    + ModifyUserPermissions(user: User, permissions:
Permission): void
}

class ArchiveStaff {
    + ManageFileInfo(file: ArchiveFile): void
    + ManageRecordInfo(record: ArchiveRecord): void
}

class BorrowManager {
    + HandleBorrowRegistration(borrow: BorrowRecord):
void
    + HandleReturnRegistration(borrow: BorrowRecord):
void
    + ProcessElectronicBorrowRequest(request:
```

ElectronicBorrowRequest): void

}

class Permission {

- ID: String

- Name: String

- Description: String

}

class ArchiveFile {

- ID: String

- Name: String

- Content: String

- Type: String

- IsDeleted: boolean

}

class ArchiveRecord {

```
    - ID: String
    - Title: String
    - Files: List<ArchiveFile>
    - IsDeleted: boolean
}

class BorrowRecord {
    - ID: String
    - File: ArchiveFile
    - Borrower: User
    - Date: Date
    - DueDate: Date
    - ReturnDate: Date
}

class ElectronicBorrowRequest {
    - ID: String
    - File: ArchiveFile
```

- Requester: User

- Date: Date

- IsApproved: boolean

- ApprovalDate: Date

}

class BackupConditions {

- Year: int

- ArchiveType: String

- IsFullBackup: boolean

}

class BackupFile {

- ID: String

- CreateDate: Date

- Content: String

- Condition: BackupConditions

}

```
class EncryptionAlgorithm {
    + Encrypt(password: String): String
    + Decrypt(encryptedPassword: String): String
}

ArchiveManagementSystem "1" *-- "1" EncryptionAlgorithm
ArchiveManagementSystem "1" -- "*" User
ArchiveManagementSystem "1" -- "*" BackupFile

User "1" *-- "0..*" Permission
User "1" -- "*" BorrowRecord
User "1" -- "*" ElectronicBorrowRequest

AdminUser --|> User
ArchiveStaff --|> User
BorrowManager --|> User
```

```
ArchiveStaff "1" -- "*" ArchiveFile

ArchiveStaff "1" -- "*" ArchiveRecord


ArchiveRecord "1" *-- "*" ArchiveFile


BorrowManager "1" -- "*" BorrowRecord

BorrowManager "1" -- "*" ElectronicBorrowRequest


ElectronicBorrowRequest "1" -- "1" ArchiveFile


BackupFile "1" -- "1" BackupConditions


@enduml
```

（22）Library management system

```
@startuml
```

```
class LibraryManagementSystem {
    + SearchBookByTitle(title: String): List<Book>
    + SearchBookByIsbnIssn(isbnIssn: String): List<Book>
}

class Borrower {
    - ID: String
    - Name: String
    - ContactInformation: String
    + GetLibraryCard(): LibraryCard
    + GetBorrowedBooks(): List<BookCopy>
}

class LibraryCard {
    - ID: String
    - BorrowerId: String
    - Date: Date
```

- ExpiryDate: Date

}

class Librarian {

- ID: String

- Name: String

+ HandleBorrow(borrower: Borrower, bookCopy: BookCopy, dueDate: Date): BorrowRecord

+ HandleReturn(borrowRecord: BorrowRecord): void

}

class SystemAdmin {

- ID: String

- Name: String

+ AddBook(book: Book): void

+ DeleteBook(bookId: String): void

+ ModifyBook(book: Book): void

+ AddBorrower(borrower: Borrower): void

```
    + DeleteBorrower(borrowerId: String): void
    + ModifyBorrower(borrower: Borrower): void
    + QueryBorrowerInfo(borrowerId: String): Borrower
    + QueryBookInfo(bookId: String): Book
    + QueryLibrarianInfo(librarianId: String): Librarian
}

class Book {
    - ID: String
    - Title: String
    - Author: String
    - ISBN: String
    - Publisher: String
    + GetCopies(): List<BookCopy>
}

class BookCopy {
    - ID: String
```

```
    - BookID: String
    - AcquisitionDate: Date
    - IsAvailable: boolean
}

class BorrowRecord {
    - ID: String
    - BorrowerID: String
    - CopyID: String
    - BorrowDate: Date
    - Date: Date
    - ReturnDate: Date
}

LibraryManagementSystem "1" -- "*" Book
LibraryManagementSystem "1" -- "*" Borrower
LibraryManagementSystem "1" -- "*" Librarian
LibraryManagementSystem "1" -- "*" SystemAdmin
```

```
Borrower "1" *-- "1" LibraryCard

Borrower "1" -- "*" BorrowRecord


Librarian "1" -- "*" BorrowRecord


SystemAdmin --|> User

Librarian --|> User


Book "1" *-- "*" BookCopy


BookCopy "1" -- "*" BorrowRecord


@enduml
```

（23）Delivery system

```
@startuml
```

```
class Person {
    - Vat
    - Name
    - Phone
}

class Customer {
    - Address
}

class Courier

class Package {
    - ID
    - Weight
    - Urgency
}
```

```
class DeliveryCenter {
    - Name
    - Address
}

class DateTime {
    - ArrivalDate
    - DepartureDate
}

Person <|-- Customer
Person <|-- Courier

Customer "1" -- "*" Package : Sender

Customer "1" -- "*" Package : Recipient
```

```
Courier "*" -- "1" DeliveryCenter

Package "*" -- "1" DeliveryCenter : Dropoff point

Package "*" -- "*" DeliveryCenter

(Package, DeliveryCenter) .. DateTime

Package "*" -- "1" Courier

@enduml
```

（24）Hi-Key Furniture Information System

```
@startuml

class Piece {
        - Reference
```

```
        - Price
}

class Line {
        - Type
}

class Component {
        - Code
        - Price
}

class ComponentType {
        - Name
}

class Order {
        - Number
```

```
        - Date
}

class Store {
        - Name
        - Address
        - FAX
}

class Quantity {
        - Quantity
}

Piece "*" -- "1" Line

Piece "*" -- "1..*" Component

(Piece, Component) .. Quantity
```

```
Component "*" -- "1" ComponentType

Piece "*" -- "*" Order
(Piece, Order) .. Quantity

Order "*" -- "1" Store

@enduml
```

（25）Factory database system

```
@startuml

class Person {
    - Name
    - Surname
    - Address
```

```
}

class Client {
    - Phone
    - ContactPerson
}

class Worker {
    - BirthDate
}

class Skill {
    - Name
}

class Machine {
    - Serial
    - Make
```

```
        - Model
        - PurchaseDate
}

class ProductType {
        - Material
        - Weight
}

class Product {
        - ID
        - Name
}

class PurchaseOrder {
        - Number
        - Date
        - Expected_Del
```

```
        - Actual_Del
}

class Quantity {
        Quantity
}


Person <|-- Client
Person <|-- Worker


Worker "*" -- "*" Skill


Client "1" -- "*" PurchaseOrder


Machine "1" -- "*" ProductType
```

```
ProductType "1" -- "*" Product

PurchaseOrder "*" -- "*" Product

(PurchaseOrder, Product) .. Quantity
Worker "*" -- "*" Machine

@enduml
```

（26）Bicycle Rental Information System

```
@startuml

class Client {
    - Nin
    - Tin
    - Name
```

```
        - Address
}

class Reservation {
        - PickupDay
        - ReturnDay
}

class Store {
        - Name
        - Address
}

class Bicycle {
        - Number
}

class BicycleModel {
```

```
        - Name
        - Type
        - No_Gears
}



Client "1" -- "*" Reservation


Reservation "*" -- "1" Store


Reservation "*" -- "1" BicycleModel


Bicycle "1" -- "*" Reservation


BicycleModel "1" -- "*" Bicycle


note right of BicycleModel
    {type should be in: road,
```

```
    mountain, bmx or hybrid}
end note

note top of Reservation
    {Return_Day >=
    Pickup_Day}
end note

@enduml
```

（27）Saturn Company Parking System

```
@startuml

class "car park"

class Barrier {
        + Raise()
```

```
}

class "card reader" {
    + Ident staff card()
    + Show success message()
    + Show fail message()
    + Eject card()
}

class "card"

class "guest card"

class "staff card" {
    + Set last access()
}

class access
```

```
class signal



"car park" o-- "2..*" Barrier


"car park" o-- "2..*" "card reader"


"card reader" o-- "*"    "guest card"


"card" "1" -- "*"    "guest card"

"card" <|-- "staff card"

"card" <|-- access


Barrier ..> signal


"card reader" o-- signal
```

@enduml

（28）OOBank Bank System

@startuml

class OrganizationalUnit

class AccountType {

- InterestRate

- MonthlyFee

- ChecksAllowed

- InsuranceAvailable

}

class Branch {

- Address

- Number

```
}

class Person {
    - Name
}

class Employee

class Customer

class Account {
    - Number
    - Balance
    - CreditOverdraftLimit
}

class ChequingAccount
```

```
class MortgageAccount {
    - CollateralProperty
}

class CreditCard {
    - Name
}

class CreditCardAccount {
    - ExpiryDate
}


OrganizationalUnit -- "*" Employee : worksFor

OrganizationalUnit "0..1" -- OrganizationalUnit : subdivision
```

```
OrganizationalUnit "0..1" -- "0..1" Employee : manager


OrganizationalUnit <|-- Branch


Branch -- "*" Account


Person <|-- Employee

Person <|-- Customer


Employee -- "*" Customer : personalBanker


Customer "1..2" -- "*" Account


Account "*" -- AccountType


Account <|-- ChequingAccount

Account <|-- MortgageAccount

Account <|-- CreditCardAccount
```

```
CreditCard "1..*" -- CreditCardAccount

@enduml
```

（29）Automobile Maintenance Information System

```
@startuml

class Person {
    - ID
    - Name
    - Address
}

class Employee {
}
```

```
class Owner {
}

class Service {
    - Car_KM
    - HoursSpent
    - NON_OEM
    - FinishDate
}

class Car {
    - Plate
    - Color
}

class Part {
    - SerialNumber
```

```
    - Price
}

class PartType {
    - Price
}

class CarModel {
}

class Make {
    - Name
}

Person <|-- Employee
Person <|-- Owner

Employee "1" -- "*" Service
```

```
Owner "1" -- "*" Car

Service "*" -- "1" Car

Service "1" -- "1..*" Part

Part "*" -- "1" PartType

Car "*" -- "1" CarModel

CarModel "*" -- "1" Make

PartType "*" -- "*" CarModel

@enduml
```

（30）Online Customer Service System

```
@startuml

class UserState {

    - New

    - Active

    - Blocked

    - Banned
```

```
}

class OrderStatus {
    - NewOrder
    - Hold
    - Shipped
    - Delivered
    - Closed
}

class "Web User" {
    - Login_id: String {id}
    - Password: String
    - State: UserState
}

class Customer {
    - ID: String {id}
```

- Address: Address

- Phone: String

- Email: String

}

class Account {

- ID: String {id}

- Billing_Address: Address

- IsClosed: Boolean

- Open: Date

- Closed: Date

}

class "Shopping Cart" {

- Created: Date

}

class Order {

- Number: String {id}

- Ordered: Date

- Shipped: Date

- Ship_to: Address

- Status: OrderStatus

- Total: Real

}

class Payment {

- ID: String {id}

- Paid: Date

- Total: Real

- Details: String

}

class LineItem {

- Quantity: Integer

- Price: Price

```
}

class Product {
    - ID: String {id}
    - Name: String
    - Supplier: Supplier
}


"Web User" "0..1" -- "1" Customer

"Web User" "1" -- "0..1" "Shopping Cart"

Customer "1" *-- "1" Account

Account "1" *-- "1" "Shopping Cart"

Account "1" *-- "*" Order
```

```
Account "1" -- "0..1" Payment

Order "1" -- "*" Payment

"Shopping Cart" "1" -- "*" LineItem

Order "1" -- "*" LineItem

LineItem "1" -- "1" Product

@enduml
```